

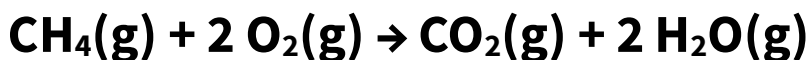
5.A Stoichiometry I

Introduction to Stoichiometry; Molar Ratios. In chemical reactions, amounts of reactants consumed and products produced are related in very specific ways. All chemical process must happen in accordance with the law of conservation of mass: chemical equations must be balanced, matter is neither created nor destroyed, etc. In other words, chemical substances will not combine in any old amount, and we need to plan out reactions quantitatively to avoid waste and ensure we'll make enough of the desired product for our purposes. Enter stoichiometry, the study of mass, volume, and molar relationships in chemical reactions. Balanced chemical equations and the number ratios they imply of different substances sit at the center of stoichiometry, and this is where we'll start.

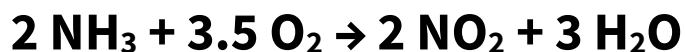
- **Stoichiometry** is the study of quantitative relationships involving mass and number of moles in chemistry.
- Any chemical question that asks “**how much?**” of a particular substance is consumed or formed in a chemical reaction is a stoichiometry problem.
- For example, consider the following pancake recipe:

1 cup mix + 0.75 cup milk + 1 egg → 8 pancakes

- Given amounts of ingredients, **how many** pancakes can we make?
- Given an amount of ingredient X, **how much** of ingredients Y and Z is needed to use up ingredient X?
- Given a target amount of pancakes we'd like to make, **how much** of each ingredient must be used?
- **Molar ratios (mole-mole ratios)** derived from stoichiometric coefficients in a balanced chemical equation are used in stoichiometry problems to “convert” from amount of one substance to amount of another.



5.1. How many moles of O_2 are required to produce 12.5 mol of NO_2 according to the balanced chemical equation below?



- Moles cannot be measured directly—in the laboratory, mass and volume measurements are typically used to quantify substances.
- **Molar mass (g/mol)** enables us to determine moles from mass or *vice versa*.

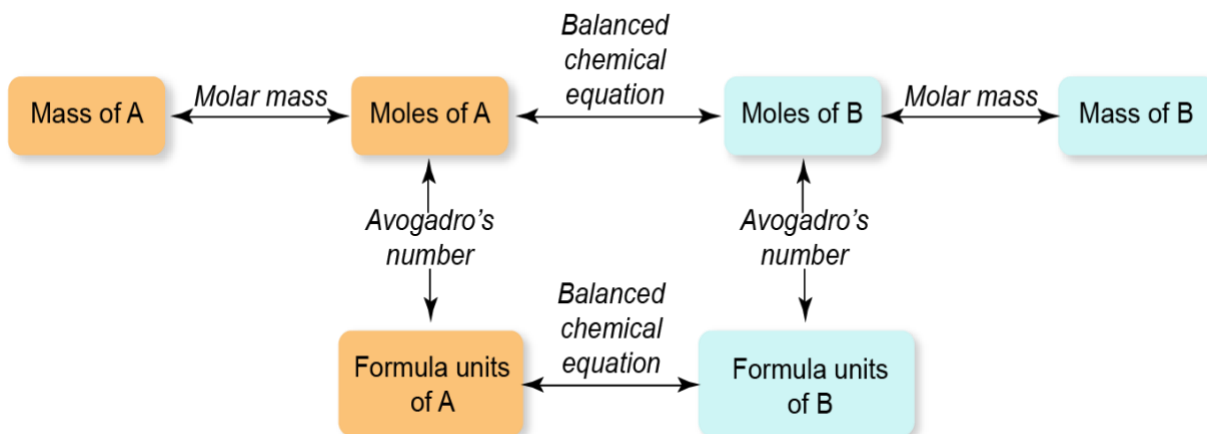
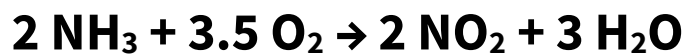


Figure 5.2. Flow chart for stoichiometry problems involving mass—the start of our stoichiometry toolbox.

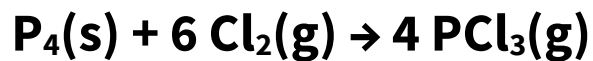
- When working a stoichiometry problem, creating a “solution flowchart” *before* plugging in numbers is highly recommended.

Example. What mass of oxygen gas (O_2) from the air is consumed in the combustion of 702 g of octane (C_8H_{18})?

5.2. How many grams O_2 are required to produce 27.5 g of NO_2 according to the reaction below?



5.3. What mass of Cl_2 is required to produce 332 g PCl_3 in the presence of [excess](#) P_4 ?



Limiting Reactant. We now understand how to figure out exactly how much of each reactant we need to use in a chemical reaction to achieve a desired yield of products. In the real world, these exact amounts are often not used—sometimes, a reaction just turns out better when an excess of one or more reactants is employed. In this case, one reactant will run out before the others; this reactant is the one that controls or *limits* the yield of products.

- The exact amount of a substance needed to consume another substance is called the **stoichiometric amount**.
- Reactants are frequently not combined in stoichiometric amounts; in this case, one reactant will run out before the others.
 - The reactant that is fully consumed is called the **limiting reactant**.
 - Reactant(s) that remain at the end of the reaction are called **excess**.

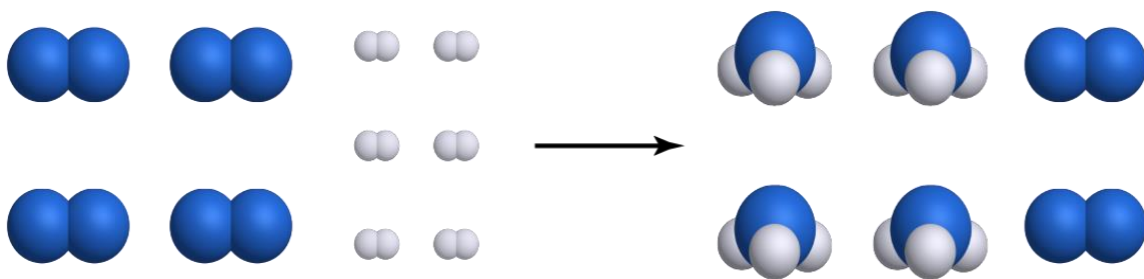
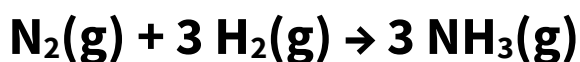
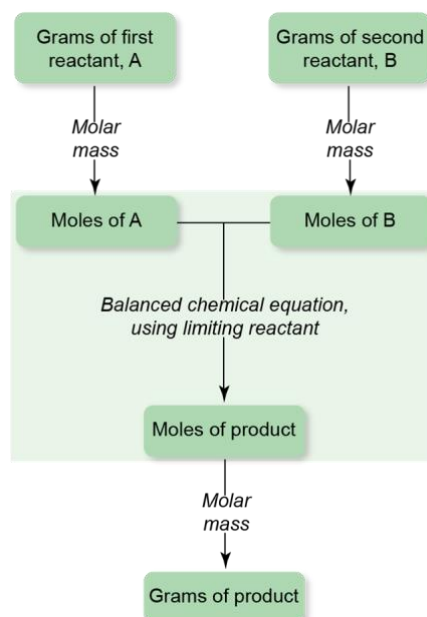


Figure 5.4. Reaction between nitrogen (blue) and hydrogen (white) to produce ammonia.

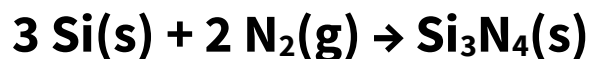
- To better understand the concept of a limiting reactant...let's make some sandwiches!
- Given the recipe below, how many sandwiches can we make with 26 bread, 20 ham, and 10 cheese?



- The limiting reactant is the substance capable of producing the *smallest* amount of product (or reaction events).
- The **theoretical mass** of product is determined *only* from the moles of the limiting reactant.
- Amounts of excess reactant(s) remaining are determined by subtracting the amount of reactant consumed from the initial amount.
- Think about sandwiches!



Example. Silicon nitride is prepared according to the following equation:

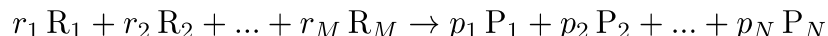


Which reactant is limiting when 2.00 g of Si and 1.50 g of N₂ react?

5.B Stoichiometry II

Limiting Reactant Continued. Now that we know how to identify the limiting reactant, we'll practice applying that knowledge to calculate theoretical amounts of products formed, excess reactant remaining, and related quantities.

- Formally, consider a reaction with M reactants R_i (with coefficients r_i) and N products P_j (with coefficients p_j), with chemical equation



- Given an initial amount in moles $n(R_i)$ for each reactant, the limiting reactant is the substance for which $n(R_i)/r_i$ is smallest. This ratio is the **number of reaction events** x_i that each reactant can carry out; the smallest x_i is the limiting number of reaction events x_{lim} .

$$x_{lim} = \min [n(R_1)/r_1, n(R_2)/r_2, \dots, n(R_M)/r_M]$$

- For any product P_j , we can calculate the **theoretical yield** in moles using

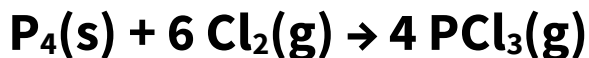
$$n(P_j) = p_j \cdot x_{lim}$$

- For each reactant that is not limiting, the excess remaining is the following, where *fin* and *init* refer to “final” and “initial” numbers of moles respectively.

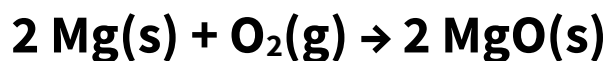
$$n(R_i)_{fin} = n(R_i)_{init} - r_i \cdot x_{lim}$$

- If this abstract math is confusing, remember...think about sandwiches!

5.4. What is the maximum mass (in g) of PCl_3 that can be produced from the reaction of 125 g P_4 with 323 g Cl_2 ?



5.5. If 42.5 g of O₂ reacts with 75.1 g of Mg to produce MgO, then what mass of excess reagent remains? Assume 100% yield for the reaction.

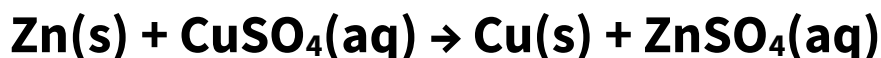


Theoretical Yield and Percent Yield. The amount in moles or mass of a product formed in a chemical reaction is known as **yield**. We're now very familiar with the idea that the law of conservation of mass constrains the yields of practical chemical reactions. We also know how to calculate theoretical yield, even in a limiting-reactant situation. A new concept we'll introduce here is percent yield, the amount of product obtained in practice as a percentage of the theoretical yield. Percent yield is an important measure of reaction quality (higher is better, provided it remains below 100%).

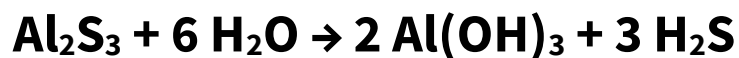
- **Theoretical yield** is the hypothetical maximum mass or moles of product given conservation of mass and a balanced chemical equation.
 - **Actual yield** is the measured mass or moles of product obtained in a practical reaction.
 - **Percent yield** is the ratio of actual yield to theoretical yield, expressed as a percentage.

$$\%Y = \frac{n_{act}}{n_{theo}} \times 100\% = \frac{m_{act}}{m_{theo}} \times 100\%$$

Example. Upon reaction of 1.274 g of copper(II) sulfate with excess zinc metal, 0.392 g copper metal was obtained according to the equation below. What is the percent yield of copper metal?



5.6. A 316.0 g sample of aluminum sulfide reacts with 493.0 g of water to produce aluminum hydroxide and dihydrogen sulfide. Determine mass of aluminum hydroxide that forms if the percent yield for the reaction is 85.0%.



Solution Concentration and Molarity. Most practical reactions take place in liquid solutions. In this context, we count solute molecules using **concentration**, a measure of the “density” of solute molecules in a solution. Using concentration, we can work from the volume or mass of a solution to the mass or amount in moles of a solute. The most important and ubiquitous measure of concentration in chemistry is **molarity**, moles of solute per liter of solution. When a molarity is known with precision (and we will soon discuss how to measure molarity precisely using titration), we can use volumetric glassware to measure a number of moles of solute with very high precision.

- Recall that in an **aqueous solution**, a **solute** is dissolved in water (**solvent**).
- Concentration** is a measure of the amount of solute in a fixed amount of solution.

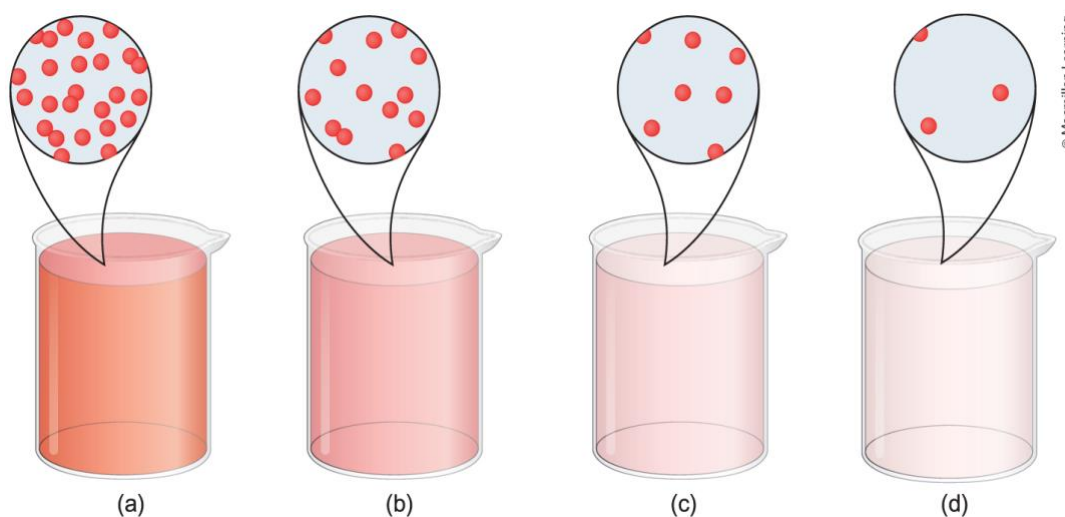


Figure 5.8. A series of solutions from concentrated (a) to dilute (d).

- The most important expression of concentration in chemistry is **molarity (mol/L, M)**.
 - Moles of solute per liter of solution
 - The molarity of a solute with chemical formula X is frequently denoted [X]. Whenever we're dealing with molarity, it's important to answer the question "moles of what?"
 - Molarity enables calculation of moles of solute from solution volume and *vice versa*.

$$[X] = \frac{n(X)}{V_{soln}}$$

$$n(X) = [X]V_{soln}$$

$$V_{soln} = \frac{n(X)}{[X]}$$

5.C Stoichiometry III

Molarities of Ionic Solutes. Dissociation of ionic solutes in aqueous solutions frequently means that the molarity of a particular ion is *not* equal to the molarity of the salt itself (e.g., $[\text{MgCl}_2] \neq [\text{Cl}^-]$). Distinguishing between the two is important. For example, we've seen before that all precipitation reactions involve irrelevant spectator ions and that we can represent precipitations using net ionic equations. The ions that appear in these equations may come from salts that contain multiple “copies” of the ions. The result is that the amount in moles of ion dissolved is not equal to the amount in moles of salt dissolved—but we really care about the former in stoichiometric analysis of precipitation reactions.

- Soluble ionic compounds dissociate completely into ions when dissolved in water. Formula units of ionic compounds may contain more than one ion!
- The “molarity of an ion” is the number of moles of that ion per liter of solution.
- Subscripts on monatomic or polyatomic ions indicate the molar ratio of that ion to the formula unit as a whole.



5.7. What is the molarity of calcium ions in a solution made by dissolving 23.26 g $\text{Ca}_3(\text{PO}_4)_2$ (0.07500 mol) in enough water to produce 200.0 mL of solution?

5.8. What mass of Na_3PO_4 is required to produce 500.0 mL of solution that is 0.750 M in *sodium ions*? The molar mass of Na_3PO_4 is 163.94 g/mol and that of Na^+ is 22.99 g/mol.

Dilution. My neighbor is a virologist at the Centers for Disease Control. In her view, the dilution equation is “the only thing those freshmen really need to learn.” Here, we’ll get comfortable determining the impact of addition of solvent to a solution. This lowers the concentrations of all dissolved solutes in a straightforward way.

- **Dilution** is the process of adding pure solvent to a solution, lowering the concentrations of all solutes.

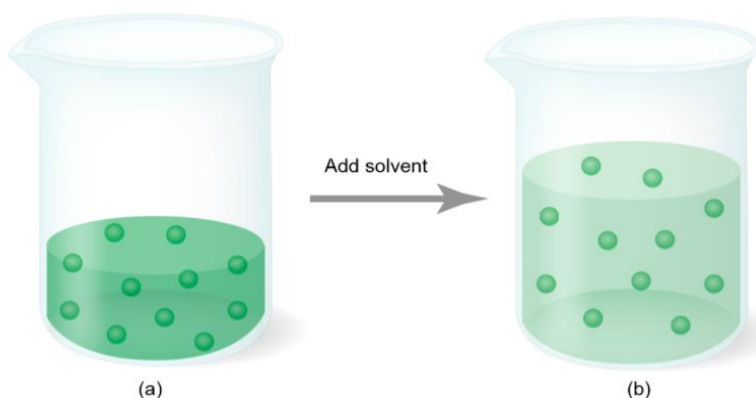


Figure 5.11. Adding solvent dilutes solution (a) to solution (b).

- The amount in moles (or mass) of solute before dilution is equal to the amount in moles (or mass) of solute after dilution. From our analysis of molarity above, we know how to calculate the moles of solute in a solution. Denoting the solution before dilution as state 1 and the solution after dilution as state 2 ($V_2 > V_1$),

$$[X]_1 V_1 = [X]_2 V_2$$

- This equation generalizes to any concentration and volume units. Units must match across the two sides of the equation, but volume and concentration units do not have to be “compatible.” For example, it’s completely fine to use molarity (mol/L) with milliliters, gallons, cubic inches, or any other volume unit in this equation.

$$C_1 V_1 = C_2 V_2$$

Example. When a 1.25 M solution of NaCl in water is diluted from 25.0 mL to 100. mL, what is the concentration in mol/L of the resulting solution?

Solution Stoichiometry. Molarity is a new tool for our “stoichiometry toolbox,” as it allows us to “convert” from solution volume to moles of solute. Like molar mass, it provides a bridge from the “laboratory world” of masses and volumes to the “mole world” in which we count molecules and think submicroscopically.

- **Solution stoichiometry** is the application of stoichiometry to reactants and/or products in solution.
- Molarity is another tool in the stoichiometry toolbox enabling us to determine moles of solute from volume of solution or *vice versa*.

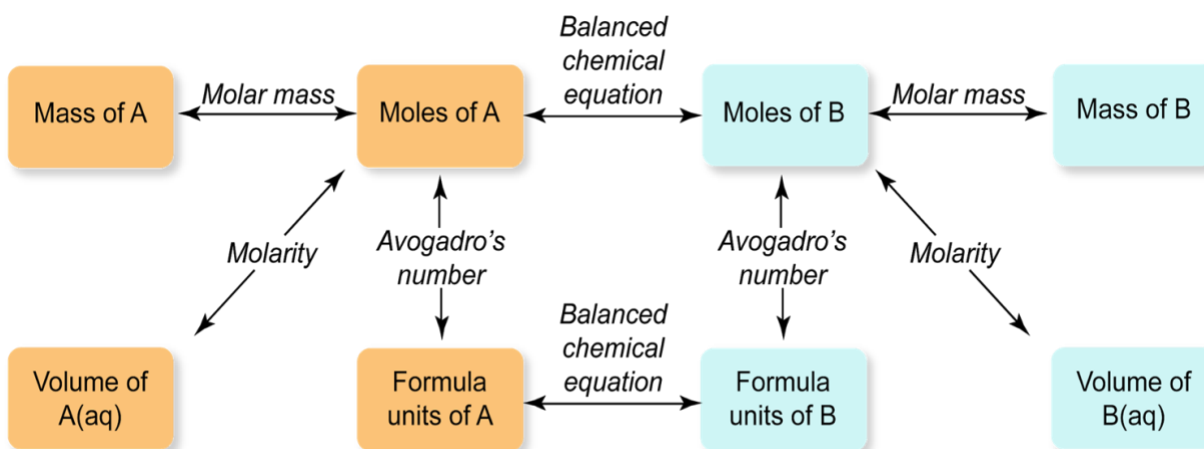
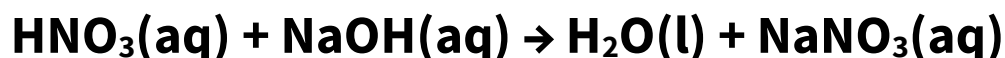


Figure 5.14. Flow chart for stoichiometry problems incorporating aqueous solutions.

5.9. A 20.00 mL sample of aqueous HNO_3 reacts with 0.110 M NaOH. Determine the concentration of HNO_3 if 49.22 mL of NaOH is required for complete reaction (with 100% yield). Enter your response to the nearest 0.001 M.



Stoichiometric Calculations Involving Other Quantities. We can realize the full power of the stoichiometry toolbox by recalling that density and Avogadro's number (and soon, the ideal gas law) can all be used to work towards counting molecules.

- Our “stoichiometry toolbox” now contains several ways to determine moles from measured quantities...
 - **Molar mass:** mass → moles
 - **Molarity:** solution volume → moles of solute
 - **Density and molar mass:** substance volume → moles
 - **Avogadro's number:** number of particles → moles
- Note that we will add the ideal gas law ($n = PV/RT$) to this list later in the course.

Example. Calculate the moles of C_2F_4 needed to produce 185 mL of CHClF_2 . The density of CHClF_2 is 1.49 g/mL.



Chapter Summary. Some of the amazing things we can now do with stoichiometry...

1. Extract molar ratios from a balanced chemical equation and use them to relate amounts of reactants and/or product to one another.
2. Relate masses of reactants and/or products to one another stoichiometrically using molar ratios and molar masses.
3. Determine limiting reactant given a balanced chemical equation and amounts or masses of all reactants.
4. Calculate theoretical yield of a given product using a balanced chemical equation and amounts or masses of all reactants.
5. Calculate or apply percent yield given an actual yield and sufficient information to determine theoretical yield.
6. Determine or apply the molarity of a given solute in an aqueous solution; apply molarity in stoichiometric calculations for reactions in solution.
7. Apply the dilution equation to plan or assess a dilution operation.